NBR-Net: A Nonrigid Bidirectional Registration Network for Multitemporal Remote Sensing Images

Yingxiao Xu, Jun Li, Chun Du, and Hao Chen

***Abstract*— Remote sensing image registration is the basis of change detection, environmental monitoring, and image fusion. Under severe appearance differences, feature-based methods have difficulty in finding sufficient feature matches to solve the global transformation and tackling the local deformation caused by height undulations and building shadows. By contrast, nonrigid registration methods are more flexible than feature-based matching methods, while often ignoring the reversibility between images, resulting in misalignment and inconsistency. To this end, this article proposes a nonrigid bidirectional registration network (NBR-Net) to estimate the flow-based dense correspondence for remote sensing images. We first propose an external cyclic registration network to strengthen the registration reversibility and geometric consistency by registering Image A to Image B and then reversely registering back to Image A. Second, we design an internal iterative refinement strategy to optimize the rough predicted flow caused by large distortion and viewpoint difference. Extensive experiments demonstrate that our method shows a performance superior to the state-of-the-art models on the multitemporal satellite image dataset. Furthermore, we attempt to extend our method to heterogeneous remote sensing image registration, which is more common in the real world. Therefore, we test our pretrained model in a satellite and unmanned aerial vehicle (UAV) image registration task. Due to the cyclic registration mechanism and coarse-to-fine refinement strategy, the proposed approach obtains the best performance on two GPS-denied UAV image datasets. Our code will be released at https://github.com/xuyingxiao/NBR-Net.**

***Index Terms*— Cycle consistency, iterative refinement, nonrigid, remote sensing image registration, reversibility, unmanned aerial vehicle (UAV) image registration.**

I. Introduction

IMAGE registration geometrically aligns images taken at different times, from different viewpoints, and/or by different sensors, which plays an increasing role in many fields, including computer vision [1]–[6], multimodal medical image alignment [7]–[10], and remote sensing image registration [11]–[14]. The massive remote sensing images come from different sources and use different coordinate systems that need registration for subsequent application as a prerequisite.

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The authors are with the College of Electronic Science and Technology, National University of Defense Technology, Changsha, Hunan 430070, China (e-mail: xuyingxiao17@nudt.edu.cn; junli@nudt.edu.cn; duchun@nudt.edu.cn; hchen@nudt.edu.cn).

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Feature-based image matching methods solve the global transformation matrix, including rotation, scale, and translation. They describe the moving image and the fixed image by feature vectors and do similarity matching based on distance or similarity metric rules. However, many algorithms, such as scale-invariant feature transform (SIFT) [15], perform well among homogeneous image matching while ineffective in heterogeneous image registration because of the vastly different appearances. Image matching focuses on extracting accurate keypoints and descriptors and filtering outer points. Nevertheless, feature vectors, detected by hand-craft methods, may contain severe outliers when dealing with images with appearance differences and large displacements, resulting in insufficient matching pairs. Even with the outlier rejection method such as the most classic random sample consensus (RANSAC) algorithm [16], such issues limit the application of image registration.

In addition, many remote sensing images have local geometric deformation, which is difficult to register by feature-based image matching. For example, mountain areas with ground fluctuation bring in complex local displacements that hinder the success of image registration. Shadow occlusions, caused by oblique shooting views or different shooting times and sun altitude angles, also produce uncertainty factors. Inspired by establishing dense correspondences between a pair of images, this article proposes a nonrigid bidirectional registration network (NBR-Net) to estimate the flow-based dense correspondence for remote sensing images with appearance difference and local deformation. We start the remote sensing image registration from multitemporal satellite images, and extend to the unmanned aerial vehicle (UAV) image registration, considering the advantages of high resolution and easy acquisition of UAV images. In the event of an emergency or GPS-denied, we hope to register the UAV images to the ortho-satellite images directly, without retraining or fine-tuning the model.

We summarize the following challenging problems for remote sensing image registration.

1) Hard to guarantee the reversibility and consistency of image registration: Suppose Image A is warped to $T(I_A)$ and is corresponded to Image B one-to-one, then $T(I_A)$ should still be warped back to Image A with one-to-one correspondence by $T(T(I_A))$, and vice versa. However, different lighting, shading, and shooting views of remote sensing images bring challenges to consistency.
2) Large appearance differences: There are significant differences between the remote sensing images, as they

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are obtained from different perspectives, times, and even sensors.

3) Local deformation: This is caused by height undulations, shadows, and different shot views. There exist severe local geometric changes and pixel distortions among remote sensing images, especially between ortho-satellite images and tilted UAV images.

Irreversibility and inconsistency are the challenging problems hindering the image registration, leading to the inaccurate flow field and distortion, especially in remote sensing image registration. To address the problems mentioned earlier, we propose an end-to-end NBR-Net to estimate the flow-based dense correspondence. The main contributions are as follows.

1) To the best of our knowledge, this article is the first to propose an external bidirectional registration network by registering Image $A \rightarrow B$ and then conversely back to Image A, and an internal iterative flow refinement strategy for remote sensing image registration. This network strengthens the registration reversibility and geometric consistency, especially dealing with images with noticeable appearance differences and large displacements.
2) To avoid content distortion and maintain local consistency, we enhance the loss constraint by introducing the content feature loss and normalized cross correlation (NCC) loss in bidirectional tasks, multistep iterations, and multiscale layers. In addition, this article introduces the semantic intersection over union (IOU) loss to punish misalignment near the buildings for improving the perceptual qualities.
3) We conduct extensive experiments on a multitemporal satellite image dataset and show the best performance superior to the state-of-the-art models. Furthermore, we extend our pretrained model to heterogeneous remote sensing image registration task and also obtain the best performance on two GPS-denied UAV image datasets, due to the cyclic registration mechanism and coarse-to-fine refinement strategy.

The rest of this article is organized as follows. Section II introduced the related works of remote sensing image registration. The proposed image registration method is detailed in Section III. Experiments and discussions are presented in Section IV. The conclusions are drawn in Section V.

II. Related Works

Image registration is to find a transformation that aligns two images of the same scene taken at different times, by different sensors, or from different viewpoints. The one that serves as a reference image is called a fixed image, and the image to be registered is called a moving image.

A. Feature-Based Matching

The feature-based approach aligns two images through the following three steps: first, a set of vectors is designed as descriptors to describe the information of interest in the two images, respectively, and then, the correct correspondence between them is determined, and the spatial transformation and position mapping are estimated accordingly. The academic community focuses on two main aspects: constructing robust descriptors and finding the correct correspondence. The first aspect emphasizes describing the invariant features in images and can be mainly classified into three categories: local feature descriptors [15], [17], global feature [18], and hybrid local + global descriptors [19], [20]. However, the hand-crafted descriptors can hardly cover the full range of information of interest. Dong *et al.* [21] trained a deep learning convolutional neural network (CNN) by creating training remote sensing image patch samples to generate a robust feature descriptor for feature matching. Zhang *et al.* [22] proposed a Siamese fully convolutional network to search high-confidence correspondence points between multimodal remote sensing images by maximizing the feature distance between positive and hard negative samples. Zeng *et al.* [23] proposed a network including the local object branch (LOB) and global semantic branch (GSB), which are used to generate the local-object-level features and global-context features, respectively. Fan *et al.* [24] proposed a framework for generating deep features via a deep residual encoder (DRE) fused with shallow features for multiresolution remote sensing image registration.

However, the massive mismatched descriptors hinder the correct matching of images. In finding the correct correspondence aspects, various feature matching methods solve the correspondence between a set of feature point sets in a high-dimensional space. The RANSAC [16] method is the most classical algorithm for filtering outliers. Bian *et al.* [25] proposed grid-based motion statistics (GMS), encapsulating motion smoothness as the statistical likelihood of a certain number of matches in a region, thus translating high match numbers into high match quality. Ma *et al.* [13] designed a guided matching strategy using the matching result on a small putative set with a high inlier ratio to guide the matching on a large putative set, which is practical for complex local distortions and removing outliers. The key idea is merely to preserve the neighborhood structures of actual potential and actual matches between two images. Yang *et al.* [26] designed a gradually increasing selection of inliers to improve the robustness of feature point registration between multitemporal remote sensing images. Zhao *et al.* [27] proposed a hierarchical network with a series of graph convolutions to extract and aggregate more reliable features from neighbors. SuperGlue [28] suppressed most outliers using graph neural network and attention to solve the matching assignment optimization problem.

B. Parametric Regression Methods

In general, feature-based methods are more robust and faster for image matching. Nevertheless, this two-stage method by constructing descriptors first and then feature matching introduces a lot of uncertainties. Instead, Rocco and Sivic [2] first proposed an end-to-end image registration approach by treating each pixel in deep feature maps as one individual feature point and incorporating a correlation layer in the CNN architecture to calculate the similarities between different feature points. The above-mentioned similarities are fed into a regression network to output the parameters of the geometric transformation. Other methods improve the registration performance by predicting richer geometric deformations

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such as homograph or thin plate spline (TPS) transformation parameters [29], using the structures-of-interest information by spatial transformer networks [30]. For remote sensing images with complex background, Li *et al.* [31] proposed a regression method using a dual DenseNet [32] model to regress the corner displacement and then apply the projection imaging equation to calculate the projection transformation matrix between the fixed and moving point coordinates. Wang *et al.* [33] constructed a two-branch regression network to map the remote sensing image and its open-source vector map into a common embedding space, and estimated 3-D transformation vector $F = [r_\theta, t_x, t_y]$ to describe the rotation and translations in the x- and y-directions. Girard *et al.* [34] corrected and updated existing cadastral maps to the corresponding satellite images by a fully CNN to search the dense displacement field. Zorzi *et al.* [35] proposed a self-supervised deep learning method for the generation of aligned and temporally coherent cadastral annotation in satellite and airborne imagery. It is challenging for the above-mentioned parameterized methods to describe geometric changes only by predicting affine or projection transformation matrix.

C. Nonrigid Registration Methods

The nonrigid registration algorithm based on optical flow prediction is flexible in dealing with local deformation. Wei *et al.* [36] proposed a small UAV-based multitemporal image registration method by multifeature guided point set registration for dynamic agricultural terrace monitoring. Dosovitskiy *et al.* [37] proposed an end-to-end CNN network to predict the optical flow field for finding dense correspondences, including learning the image feature representations and matching them at different locations. Wei *et al.* [36] designed the multifeature descriptor and then modeled the nonrigid displacement function by requiring it to lie within a specific functional space. Yang *et al.* [38] introduced a geometric constraint term into the L2-minimizing estimate (L2E)-based energy function for better behaving the nonrigid transformation. Global-Local Universal Network (GLU-Net) [6] proposed a universal dense correspondence network and integrated global and local correlation layers to handle large and small displacements. Zhu *et al.* [39] proposed a cycle-consistent generative adversarial network (GAN) by training two mapping generators $G : X \rightarrow Y$ and $F : Y \rightarrow X$ and two adversarial discriminators D_X and D_Y. Mahapatra *et al.* [40] also used GANs like cycleGAN to register images with the deformation field. VoxelMorph [9], BIRNet [41], DeepAtlas [42], GraphRegNet [43], etc. performed medical image registration by supervised, semisupervised, or unsupervised learning to generate distorted images and introduced the grayscale between images into measuring the image similarity as the loss function. However, the generation model is tricky to train and converge for remote sensing images, and produces distortion in image content. Mok and Chung [44] proposed to learn the two separated time 0.5 deformation fields that warp both Image A and mono-modal Image B to their mean shape M in the geodesic path. Arar *et al.* [45] performed multimodal natural image registration using two networks, including a spatial transformation network learning the geometric transformation required for the registration task and a translation network only performing for photo-metric mapping. However, those methods based on gray similarity are challenging to apply the geomatics field to register multitemporal remote sensing images. Unsupervised methods are premature for remote sensing image registration with large distortion and appearance changes.

For the image registration, the moving Image A that is warped to $T(I_A)$ should be registered to the fixed Image B, and transformed image $T(I_A)$ is warped to $T(T(I_A))$, which should be inversely registered to the moving Image A. Here, $T()$ represents the warp function for the moving image. In our opinion, this is the most important problem that hinders the improvement of image registration performance. The experimenter defines the moving and fixed images, and the one with calibration and clarity is usually regarded as the fixed one. However, the fixed image and moving image should be able to convert to each other and have reverse consistency. We tried to train an end-to-end neural network to guarantee cyclic consistency in the bidirectional processes starting from the supervised manner. We detail the bidirectional strategy to ensure the cyclic consistency in Section III.

Ma *et al.* [46] introduced a coarse-to-fine strategy and developed a two-step registration method based on deep and local features. The first step designed a CNN to calculate the approximate spatial relationship. The second step applied a matching strategy to the local feature-based method. Krebs *et al.* [47] proposed an iteration-based nonrigid image registration technique by a deep regression network to learn how to modify the spatial transformation parameters. Goforth and Lucey [48] iterative refined the image alignment matrix using a joint optimization between UAV image and satellite image. Minallah and Ashfaq [49] ran the Demon algorithm multiple times to improve the performance. Papadomanolaki *et al.* [50] proposed a deep-learning-based multistep deformable registration scheme for aligning pairs of satellite imagery. However, the image distortion is only several pixels, and only two images are cropped to train and test. In summary, it is hard to optimize for remote sensing image registration, especially when dealing with large displacements.

Another important feature for remote sensing image registration is its complex background and details. It is hard to align every pixel, like a set of corresponding points on a lawn. We hope that the nonrigid registration network can automatically learn smooth optical flow field like perspective transform and infer the registration relationship of the area with appearance change from the area with higher registration probability. Moreover, we are very concerned about areas with significant semantic features, such as buildings, roads, and rivers. This semantic information plays an essential role in guiding registration. Semantic attribute matching network (SAM-Net) [51] presented a semantic attribute matching loss based on the matching similarity between an attribute transferred source feature and a warped target feature in the face domain adaptation. Huang *et al.* [52] designed an end-to-end learnable deep network, named dynamic context correspondence network (DCCNet), to incorporate global semantic context flexibly to overcome the limitation that relies on local

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semantic representations. Semantic GLU-Net [6] added a consensus network for the global correlation layer and concatenated features from different levels. All the above-mentioned approaches exploit the semantic information in the images and take promising results. The semantic information in remote sensing images can also be represented more explicitly. This article uses remote sensing images' accessible semantic building labels as constraint terms to further enhance the image alignment effect.

III. Proposed Method

A. Problem Definition

In this article, we explore how to register remote sensing images by predicting the pixelwise flow field from the moving image $I_A \in \mathbb{R}^{H\times W\times 3}$ to the fixed image $I_B \in \mathbb{R}^{H\times W\times 3}$. Our goal is to estimate a dense displacement flow field $F_{A\to B} \in \mathbb{R}^{H\times W\times 2}$ that warps image I_A toward I_B such that

$$I_B(x) \approx T(I_A) = I_A(x + F_{A\to B}(x)). \tag{1}$$

The image $T(I_A)$ represents the warped Image A according to the predicted flow field $F_{A\to B}$. The flow $F_{A\to B}$ represents the pixelwise 2-D motion vectors in x- and y-coordinates of the target image to its corresponding position in the source image.

1) Bidirectional Consistency: The image I_A is converted to image $T(I_A)$ according to the flow field $F_{A\to B}$, where $T(I_A)$ is one-to-one aligned with I_B. Then conversely, image $T(I_A)$ should be converted back to image I_A by the flow field $F_{B\to A}$, where $T(T(I_A))$ is one-to-one aligned with I_A. Here, the flow field $F_{B\to A}$ is the opposite number of values for each element of the flow field $F_{A\to B}$. That is

$$I_A(x) \approx T(T(I_A)) = T(I_A)(x + F_{B\to A}(x)). \tag{2}$$

Realizing the remote sensing image registration with different appearances through the domain adaption or generation models will cause severe image distortion or content distortion. Here, we design a bidirectional registration strategy by one network, instead of content transformation, to ensure geometric consistency and reversibility, even with the complicated elements in remote sensing images. At the same time, the stricter bidirectional registration enforces the smoothness of the optical flow field.

2) Iterative Refinement: Considering the large displacement between two images, the network is iteratively refined to enhance the flow-field estimation performance. In the refinement iterations, the previous round's fixed image and warped image are used as input. Taking $I_A \to I_B$ as an example, assume that I_A is warped to $T^1(I_A) = I_A(x + F^1_{A\to B}(x))$ in the first round transformation, we have the transformed image $T^2(I_A)$ in the second round transformation

$$I^2_B(x) \approx T^2(I_A) = T^1(I_A)\big(x + F^2_{A\to B}(x)\big). \tag{3}$$

The deformation flow field $F^k_{A\to B}, k = 1, 2, \ldots$ represents the flow field in the k refinement round. Each predicted flow in the iterative process is integrated into the loss function for guiding the network training feedback.

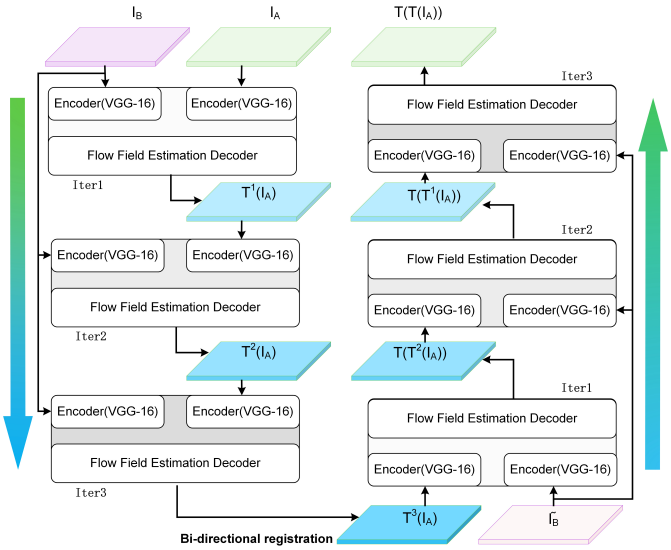


Fig. 1. Overview of the proposed method NBR-Net for remote sensing image registration, taking three internal iterations as an example. As the network predicts flow iteratively, the color darkens. The Image $\tilde{I}_B = I_B(x + F^{GT}_{B\to A}(x))$ means the GT warped image $T^{GT}(I_B)$, aligning with Image A while remaining the content style with Image B.

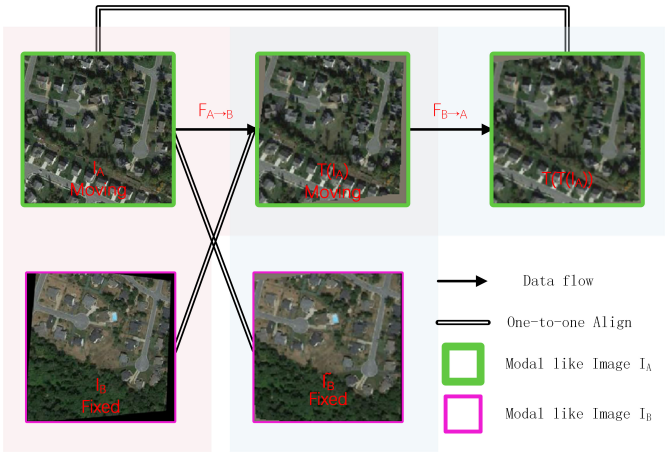


Fig. 2. Data flow. Image $T(I_A)$ is aligned with Image B. Image A, $T(T(I_A))$, and $\tilde{B}$ are one-to-one aligned. Image B and $\tilde{B}$ are the same modal, and other images are the same modal.

B. Overall Network Architecture

The overview of the structure is shown in Fig. 1.

1) Bidirectional Registration: Regarding Image A as the moving image and Image B as the fixed image, we get the warped image $T(I_A)$ aligning with Image B by the predicted flow $F_{A\to B}$. Conversely, if we regard the warped image $T(I_A)$ as the moving image and Image $\tilde{B}$ as the fixed image, the network is also expected to estimate the flow field $F_{B\to A}$ to get the warped image $T(T(I_A))$ aligning with Image A. Fig. 2 details the data flow during training; only the first step with light pink background is used for testing. In terms of correspondence, I_B and $T(I_A)$ are one-to-one aligned, and I_A, $T(T(I_A))$, and $\tilde{I}_B$ are one-to-one aligned. As for the image style, I_A, $T(I_A)$, and $T(T(I_A))$ are the same. Two registration tasks (from I_A to I_B and from $T(I_A)$ to $\tilde{I}_B$) are from the source modal to the target modal. In this way, the influence of style

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changes on the network is reduced. The network only needs to learn the corresponding relationship of pixels, and does not need to consider the corresponding relationship between styles. However, learning style mappings, including textures and colors, can be challenging while learning registration tasks online. Different from the other works' handling domain difference via domain adaption, bidirectional registration remains the image style and strengthens the consistency. Unlike the works using GANs training two independent generators and discriminators, we employ the same network and share parameters in the second step. The flow field $F_{A\to B}$ and $F_{B\to A}$ are equal to zero while both of them are learned from the same network, resulting in the effectiveness of the cycle consistent. Even with different appearances, this model also works well when aligning remote sensing images. The cycle consistency implied in the flow field should also work in other dense correspondence scenarios.

2) Iterative Refinement: For large-distortion remote sensing image registration, dense correspondence using a single step image registration may be hard to optimize or inaccurate. We adopt iterative refinement to make predictions of the flow field for each step, in each direction. Like [50], we iterate three times to give a tradeoff between accuracy and time complexity. Taking three internal iterations in Fig. 1(a), the first iteration takes Images A and B as input, and outputs the first warped image $T^1(I_A)$. The second iteration takes Image B and the previous warped image $T^1(I_A)$ as input, and outputs the warped image $T^2(I_A)$. The second predicted flow field $F^2_{A\to B}$ is equal to $F^1_{A\to B}$ and the predicted flow field in the second round—similarly hereinafter.

Each iteration step comprises two encoder modules and a flow-field estimation decoder called an NBR-Net unit.

3) Encoder Module: The encoder module extracted features by sequential convolution operations from the fixed image and the moving image separately. To be detailed, we use the VGG-16 [53] network pretrained on ImageNet. The parameters of this part network are fixed during training and testing to provide a fair comparison with [5] and [6]. Another reason to use the frozen VGG-16 for feature extraction is that we expect the models only training the decoder module can be extended to the registration between ortho-satellite remote sensing image and tilted UAV images with appearance differences. The decoder module inputs the extracted features from Conv5-3 layer (16×16 resolution) and Conv4-3 layer (32×32 resolution) and outputs the predicted flow field for the moving image.

4) Decoder Module: First, the correlation layer calculated the similarity between each position of two feature vectors to evaluate the matching probability. Given the feature vectors f_A and f_B extracted from Images I_A and I_B, the outputted correlation map $c_{ijkl} \in R^{H\times W\times(H\times W)}$ contains the scalar product of a pair of individual descriptors

$$c_{ijkl} = f_B(k,l)^T f_A(i,j) \tag{4}$$

where (i, j) and (k, l) indicate the individual feature positions in the feature vectors. Due to the large distortion and the complicated background in the remote sensing image, the allocation of matching relation among the nearest neighbors will lead to a large number of false matching. We adopt the consensus idea [54] to reduce the initial inconsistent matches. The softer version of the mutual nearest neighbor filtering $M()$ is both in the sense of softer decision and better differentiability properties that can be applied on dense 4-D match scores

$$\hat{c_{ijkl}} = r^A_{ijkl} r^B_{ijkl} c_{ijkl} \tag{5}$$

where r^A_{ijkl} and r^B_{ijkl} are the ratios of the score of the particular match c_{ijkl} with the best scores along each pair of dimensions corresponding to Images A and B, respectively

$$r^A_{ijkl} = \frac{c_{ijkl}}{\max_{ab} c_{abkl}} \tag{6}$$

$$r^B_{ijkl} = \frac{c_{ijkl}}{\max_{cd} c_{ijcd}}. \tag{7}$$

Then, we apply the consensus module $N()$ twice for c_{ijkl} and c^T_{ijkl} to produce matches whether an image pair (I_A, I_B) or (I_B, I_A) is the input to the net. The module is comprised of a sequential convolution layer and Relu. With the consensus module filtering out those negative correlations, a sequential convolution layer is applied to convert it to the predicted flow field normalized to $[-1, 1]$, and then the Warp layer outputs the $H \times W \times 2$ deformable flow field according to the correlation between two feature maps. According to the predicted optical flow, the Warp layer transforms the moving image's feature in each resolution instead of learning the remapping parameters in the network. Followed by [6], we also employed four decoders for resolution 16*16, 32*32, and $H/8 * W/8$ and $H/4 * W/4$ pyramid feature resolutions, respectively. The final NBR-Net unit outputs the predicted flow $F_{A\to B}$. Table I details the coarsest resolution's operation of the flow-field estimation decoder in the NBR-Net unit.

TABLE I
ONE FLOW-FIELD ESTIMATION DECODER IN THE NBR-NET UNIT

Moving Image A		Fixed Image B	
L4 (32×32)	Up(L5) (32×32)	L4 (32×32)	Up(L5) (32×32)
Correlation: according to Eq. (4)			
Mutual Matching: according to Eq. (5) Eq. (6) Eq. (7)			
Neighbourhood Consensus Module: N(c) + $(\mathrm{N}(\mathrm{c}^T))^T$			
Mutual Matching: according to Eq. (5) Eq. (6) Eq. (7)			
Flow Estimation Decoder: 3*3 /128 + BN + Relu			
Flow Estimation Decoder: 3*3 /128 + BN + Relu			
Flow Estimation Decoder: 3*3 /96 + BN + Relu			
Flow Estimation Decoder: 3*3 /64 + BN + Relu			
Flow Estimation Decoder: 3*3 /32 + BN + Relu			
Warp: grid_sample function			

C. Loss Function

Given the image pair to be registered, our goal is updating the network parameters to produce accurate and robust transformation field flow. It is achieved by minimizing the loss between the predicted flow field and the ground-truth (GT) flow field. In this article, we apply supervision in bidirectional

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tasks, multistep iterations, and multiscale layers, including the endpoint error (EPE) loss with the GT displacements, the intersection of union loss with the GT building labels, the content feature loss, and the NCC loss between the warped image and target fixed image.

1) EPE Loss: EPE, the standard error measure in flow estimation, is the mean of the predicted Euclidean distance between the predicted and the GT optical flow field across all pixels. Taking $A \rightarrow B$ as example, we have EPE loss

$$L_{\text{EPE}} = \sqrt{\frac{1}{m}\sum_{i=1}^{m}\left(F_{A\rightarrow B} - F_{A\rightarrow B}^{GT}\right)^2}. \tag{8}$$

2) IOU Loss: Landmarks such as roads and buildings in remote sensing images play an important role in alignment, which should be paid more attention to the alignment effect of pixels around the artificial buildings. We download the semantic building labels from OpenStreetMap (https://www.openstreetmap.org) and explicitly enhance the penalty in the vicinity of landmarks with semantic building labels. That is, the label of Image A L_A should be aligned with the label of Image B L_B according to the predicted $F_{A\rightarrow B}$. Equations (9) and (10) represent the semantic constraint between the labels of two images L_A and L_B

$$L_B(x) \approx T(L_A) = L_A(x + F_{A\rightarrow B}(x)) \tag{9}$$

$$L_A(x) \approx T(L_B) = L_B(x + F_{B\rightarrow A}(x)). \tag{10}$$

Taking $A \rightarrow B$ as an example, we have IOU loss as follows, where $T(L_A)$ represents the warped label of Image A

$$L_{\text{IOU}} = \frac{L_B \cap T(L_A)}{L_B \cup T(L_A)}. \tag{11}$$

To avoid the warped image's content distortion, we employ the content feature loss and NCC loss to constrain.

3) Feature Loss: We evaluate the feature consistency using the VGG-16 model pretrained on ImageNet to extract features for ensuring the consistency of bidirectional registration tasks. Considering registration of Image A to Image B, the features of warped image $T(I_A)$ should be aligned with Image B. We have feature loss

$$L_{\text{feat}} = \sum_{j=1}^{5}(\text{Feat}_j(T(I_A)) - \text{Feat}_j(I_B)) \tag{12}$$

where j represents the pyramid level of VGG features.

4) NCC Loss: The correlation layer in the network matches two features in different positions by measuring the correlation between two images. Here, to measure the correlation, especially the local correlation, we use the NCC loss as our similarity metric to compute the degree of local alignment between the warped image $T(I_A)$ and the fixed image I_B. Taking $A \rightarrow B$ as an example, let I and J be two input image patches, $\overline{I(x)}$ and $\overline{J(x)}$ be the local mean of I and J at position x, respectively. The local mean is computed over a local w^2 window centered at each position x, with $w = 7$ in our experiments. We have NCC loss

$$L_{\text{NCC}} = \sum_{x\in\Omega} \frac{\sum_{x_i}(I(x_i) - \overline{I(x)})(J(x_i) - \overline{J(x)})}{\sqrt{\sum_{x_i}(I(x_i) - \overline{I(x)})^2 \sum_{x_i}(J(x_i) - \overline{J(x)^2})}} \tag{13}$$

TABLE II
DATA DISTRIBUTION

		Total	Affine	TPS	Homo
Train	Samples	1600	531	525	544
	Mean	11.44	19.54	7.36	7.47
	Max	104.47	104.47	24.74	34.28
Test	Samples	400	130	133	137
	Mean	11.63	20.24	7.59	7.38
	Max	95.08	95.08	22.72	31.74



Fig. 3. Typical multitemporal remote sensing images. This dataset contains many images from both urban locations and low-texture rural locations.

where x_i denotes the position within w^2 local windows centered at x. In addition, the total loss can be written as

$$\text{Loss} = \sum_{k=1}^{3}\sum_{j=1}^{5}\sum_{i=1}^{2}(L_{\text{EPE}}^{i,k} + \lambda_1 L_{IOU}^{i,k} + \lambda_2 L_{\text{feat}}^{i,j,k} + \lambda_3 L_{\text{NCC}}^{i,k}) \tag{14}$$

where i represents the bidirectional registration tasks, j represents the pyramid level of features, and k represents the number of iterations. In this article, we believe that these loss functions play an equally important role in improving the overall registration performance. While the value ranges of these losses are different, we unify them to the same order of magnitude through coefficients. In the experiment, $\lambda_1 = 1e^4$, $\lambda_2 = 1e^-3$, and $\lambda_3 = 1e^3$.

IV. Experiments and Discussions

A. Datasets

1) Multitemporal Satellite Image Dataset: We collected multitemporal satellite images during different seasons from 2015 to 2017, covering a 27.52 km by 22.016 km geographic area in Virginia, USA. There are six large (25 600 × 20 480) images in level 18. These multitemporal remote sensing images are divided into two categories: one for Class $\mathbf{A} = \{A_1, A_2, A_3, A_4\}$ for the fall and winter seasons and the other for Class $\mathbf{B} = \{B_1, B_2\}$ for the spring and summer seasons.

We randomly select two multitemporal images from Class **A** and Class **B**, crop to 256 × 256 size image patch, and randomly generate the transformations of affine (six parameters), perspective (eight parameters), and TPS (18 parameters) like [2], [5], and [6]. One patch is transformed into a moving image according to the generated parameters. No matter the rigid or nonrigid transformation, we employ the same registration network to predict the flow field. Table II shows data distribution in the train and test datasets, including the number of samples, mean displacement pixels, and max displacement pixels. Fig. 3 shows typical examples of multitemporal image

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registration. Note the differences such as seasonal variation of vegetation, shadow angles, and perspectives of buildings.

2) GPS-Denied UAV Image Datasets: UAV image localization may not be available in GPS-denied or GPS-spoofed situations. An auxiliary vision-based system registering the UAV image and the existing satellite image could be beneficial when there are signal issues. We collected two UAV image datasets and their corresponding satellite images and evaluated the performance when extending to registration between the tilted UAV images and the ortho-satellite images.

1) *Village Dataset:* The Village dataset is proposed in [48], which is captured overhead the Swiss village of Merlischachen, at an altitude of 0.2 km, over 0.85 km flight distance. This is an urban environment with ample texture and landmarks such as buildings and roadways. The UAV images were captured in April 2013, and the satellite image in May 2012, downloaded from Google Earth Pro.
2) *Hunan Dataset:* The Hunan dataset is taken in May 2018, from ShaoYang, Hunan Province, China, with an average altitude of 630 m and an area of about 5 km^2, covering villages, factories, mountains, and farmlands, including 500+ buildings with different architectural styles. The satellite image is obtained from the open-source map from Google Map (2015). The initial position offset of UAV images in the Hunan dataset was slightly larger, reaching about 30 m.

B. Baselines

We re-implement these state-of-the-art approaches for image registration comparisons. Main information of these methods is summarized as follows:

1) *GeoCNN [2]:* GeoCNN estimated geometric transformation parameters between two input images by regression network, including feature extraction, matching and inlier detection, and model parameter estimation.
2) *DGCNet [5]:* Dense Geometric Correspondence Network (DGCNet) represents input image pairs by a Siamese neural network with shared weights and established dense pixel correspondences between images. DGCNet also designed a matchability branch to output a probability map to indicate a confidence score for each pixel location in the predicted flow.
3) *GoCor [55]:* To resolve matching ambiguities between feature vectors extracted from pairs of locations, GoCor proposed a correspondence volume by learning spatial matching priors to optimize the matching process.
4) *GLU-Net [6]:* GLU-Net proposed a dense correspondence network and integrated global and local correlation layers to handle large and small displacements. In terms of semantic correspondence, Semantic GLU-Net (Sem GLU) added a consensus network for the global correlation layer and concatenated features from different levels.

First, we evaluate the image registration performance on the multitemporal and large-distortion remote sensing dataset, and compare these models from Class **A** to **B** and from Class **B** to **A** separately. CNNGeo uses different models to evaluate affine, homograph, and TPS transformation. Therefore, we train and test this model on three subsets for three types and calculate the mean EPE error. Table III shows quantitative results in EPE and IOU metric. Experiments demonstrate that NBR-Net obtains the smallest EPE error and the highest IOU score on both registration tasks. Our NBR-Net achieves 1.8546 and 0.8965 pixels improvement than the Sem GLU approach in EPE metric, and 3.78% and 1.87% higher in IOU metric.

TABLE III
QUANTITATIVE COMPARISONS WITH OTHER BASELINES (BOLDED: BEST; UNDERLINED: RUNNER-UP)

	EPE		IOU		Time(s)
	A $\rightarrow B$	B $\rightarrow A$	A $\rightarrow B$	B $\rightarrow A$	
GeoCNN[2]	15.4490	13.8571	-	-	0.0162
DGCNet[5]	7.4487	6.8435	0.7213	0.7443	0.0329
GLU-Net[6]	7.9687	7.5318	0.7557	0.7608	0.0515
GoCor [55]	6.5746	7.2830	0.7618	0.7504	0.1363
Sem GLU[6]	<u>5.3382</u>	<u>5.8067</u>	<u>0.8067</u>	<u>0.7998</u>	0.1549
NBR-Net	**3.7399**	**4.9102**	**0.8384**	**0.8150**	0.3871

Qualitative results are shown in Figs. 4 and 5. Fig. 4 shows the registration from Class A to Class B, including four typical scenarios: low texture, roads, buildings, and large deformation. In the first example, roads are large-distorted to be registered. The DGCNet registration result is blurred, and the roads are zigzagged caused by the insufficient smoothness of the predicted optical flow. The optical flow predicted by GLU-Net, GoCor, and Sem GLU is relatively smooth, but the registration results marked in red box are not satisfactory. Our method NBR-Net improves the registration results around the distorted parts and has a good consistency. The second example is a low-textured playing field. From observation, only half of the building in the lower-left corner, a white building on the right (marked in red boxes), and the two irregular yellow playing fields in the image seem to be noticeable. Our method has achieved an excellent registration effect in these places, and the registration effect of Sem GLU is also acceptable, which can be attributed to the latent semantic information. The long buildings in the third example demonstrate the importance of explicitly introducing semantic information, which significantly reduces the deformation of the building. The two images in the fourth example have noticeable appearance changes. The roads and buildings in the lower-left corner of the moving image do not exist in the fixed image. Our method also obtained good registration results based on the partial corresponding information. This may be attributed to the smooth predicted flow field similar to perspective transformation, where the flow field in the areas with appearance changes can be inferred according to the registration relationship at other positions. In addition, the stricter geometric reversibility and consistency of bidirectional registration also enhance this performance.

Similarly, the roads in the first and second examples and the buildings in the third example in Fig. 5 also achieved the best results than the state-of-the-art baselines. Thus, our approach NBR-Net demonstrates competitive results for multitemporal and large-distortion remote sensing image registration tasks,

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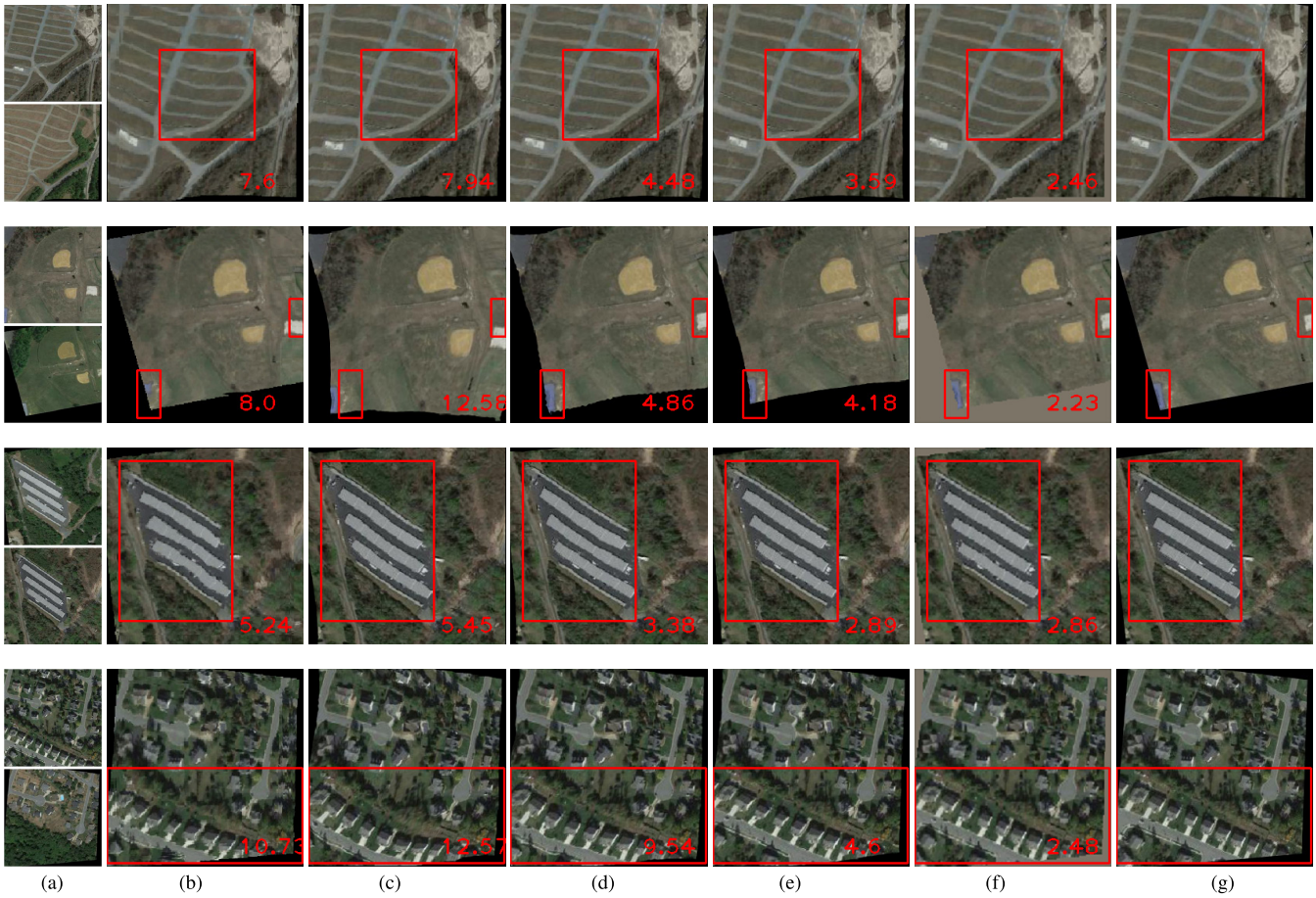


Fig. 4. Illustration of registration performance of baselines from Class A to Class B. From left to right: Unregistered image pair (top: moving; bottom: fixed), DGCNet, GLUNet, GoCor, Sem GLU, our NBR-Net, and GT. The EPE error is marked in the figure, the smaller the better. (a) Un-R. (b) DGCNet. (c) GLU-Net. (d) GoCor. (e) Sem GLU. (f) NBR-Net. (g) GT.

mainly attributed to the bidirectional consistency constraints and iterative refinement strategy.

C. Ablation Study

This section conducts comprehensive ablation experiments to demonstrate how the proposed bidirectional consistency, iterative refinement, and loss functions influenced image registration results. Quantitative comparisons with the ablation models are exhibited in Table IV. Note that the basic EPE loss is employed in every ablation model. The "semantic" in the third column represents that the network is trained with the IOU loss, and the "content" in the fourth column represents that the network is trained with the content feature loss and NCC loss. Our NBR-Net performs best and gains 29.94% and 12.92% improvement in EPE metric in the $A \rightarrow B$ and $B \rightarrow A$ tasks. Our ablation model only with the bidirectional consistency improves the registration performance by 10.81% and 11.29%, respectively. With the iterative refinement, the registration performance improves by 7.20% and 3.83%. With the explicitly semantic information constraint, the registration performance improves by 7.20% and 6.42%. In addition, the multiple loss functions improve the registration performance by 10.69% and 7.36%, respectively.

TABLE IV
QUANTITATIVE COMPARISONS WITH ABLATION MODELS
(BOLDED: BEST; UNDERLINED: RUNNER-UP)

Model	Cycle	Iter	IOU	Cont	A $\rightarrow B$	B $\rightarrow A$	Time(s)
1					5.3382	5.8067	0.1549
2			✓		4.9536	5.4339	0.1549
3			✓	✓	4.7673	5.3795	0.1549
4		✓			4.9539	5.5843	0.2810
5		✓	✓	✓	4.8007	5.4587	0.2810
6	✓				4.7612	5.1513	0.1553
7	✓		✓	✓	<u>4.0547</u>	<u>5.0562</u>	0.1553
8	✓	✓	✓	✓	**3.7399**	**4.9102**	0.3871

1) Bidirectional Consistency: Fig. 6 shows two registration examples of bidirectional consistency by 7×327 chessboard. View from the seam lines in the images. The example in the first row aligns Image B in Class B to Image A in Class A, and the second example shows registration from subset A to subset B. The registration errors are reduced to 2.68 and 8.41 only with the bidirectional consistency. By one network, the multitemporal image registration task improves the cyclic consistency constrained by two steps: Image B (source modal) is transformed to image $T(I_B)$ (source modal), which is aligned with Image A (target modal), and then

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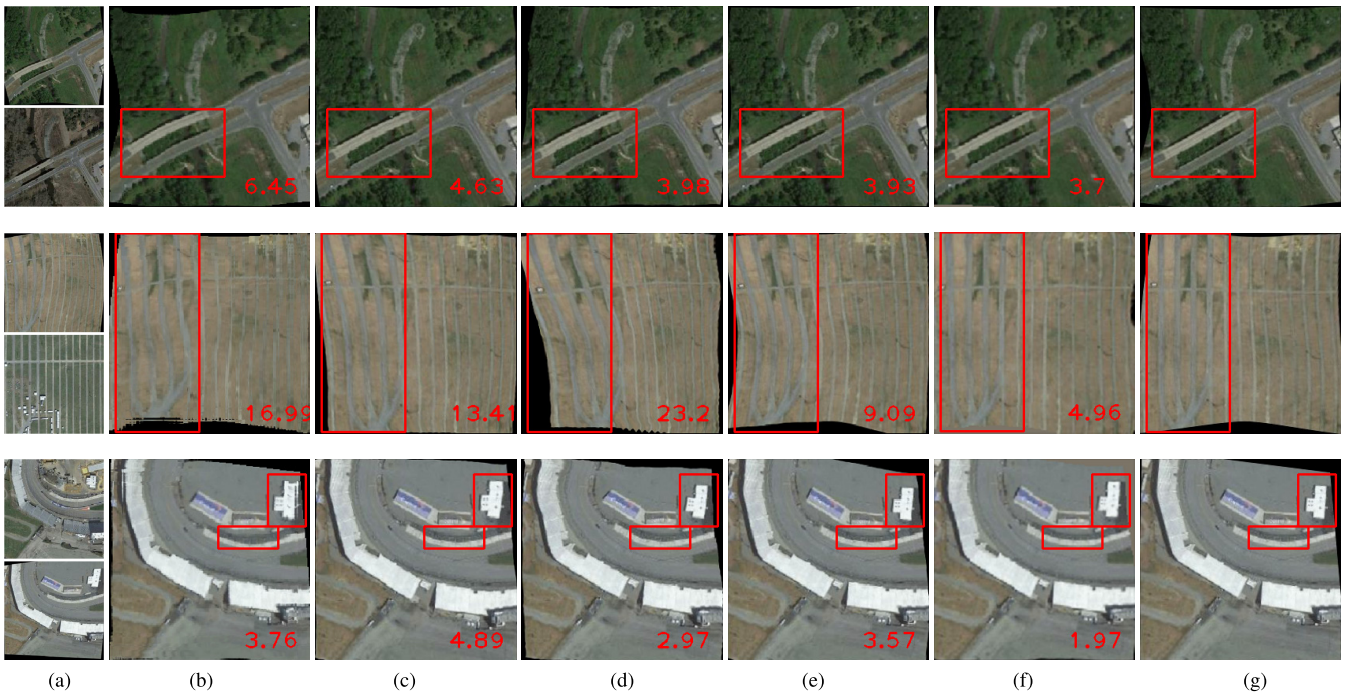


Fig. 5. Illustration of registration performance of baselines from Class B to Class A. Unregistered image pair (top: moving; bottom: fixed), DGCNet, GLUNet, GoCor, Sem GLU, our NBR-Net, and GT. The EPE error is marked in the figure, the smaller the better. (a) Un-R. (b) DGCNet. (c) GLU-Net. (d) GoCor. (e) Sem GLU. (f) NBR-Net. (g) GT.

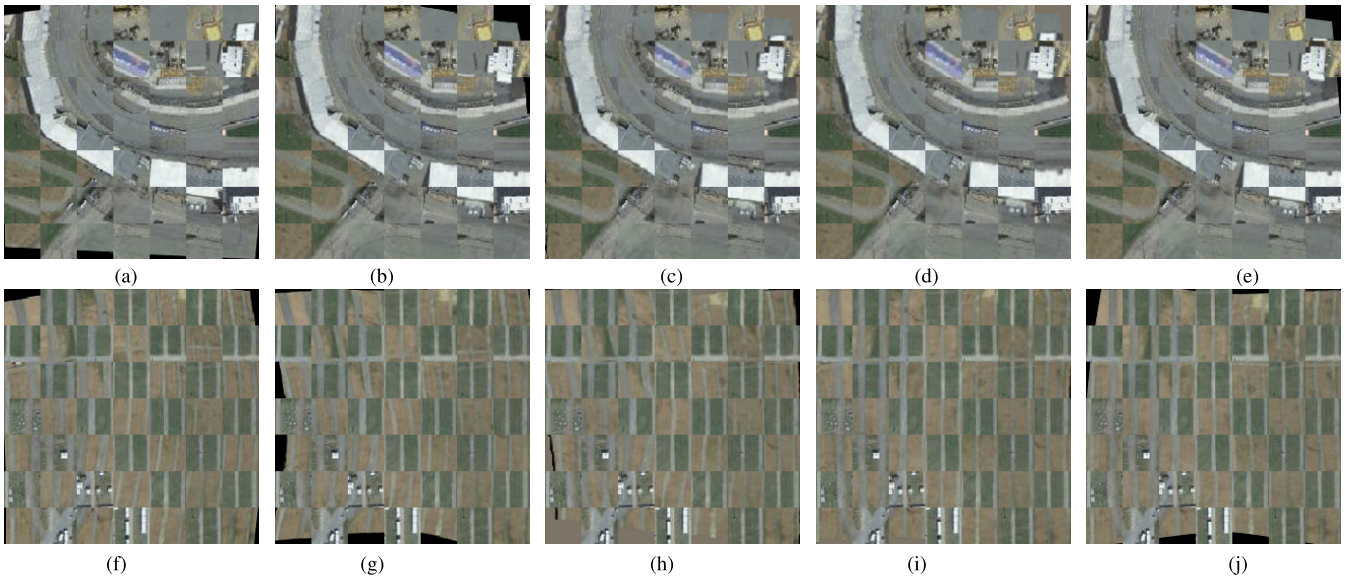


Fig. 6. Visualization of the effect of bidirectional consistency. (a) Unregistered. (b) Sem GLU: 3.57. (c) Cycle: 2.68. (d) NBR-Net: 1.97. (e) GT. (f) Unregistered. (g) Sem GLU: 9.09. (h) Cycle: 8.41. (i) NBR-Net: 4.96. (j) GT.

$T(I_B)$ is reversely transformed back to the image $T(T(I_B))$ (source modal), which is aligned with image $\tilde{A}$ (target modal). Caused by this key idea, the mono-directional multitemporal image registration is converted to a bidirectional registration task, improving the consistency and reversibility rigorously and avoiding the fault by content style transfer.

2) Iterative Refinement: It is difficult to deal with the scene with large displacement change in single-step optimization, and the iterative operation enhances the registration consistency strictly by multiple steps. Iterative refinement improves the mathematical consistency of registration transformation. The iterative refinement, semantic loss, and content loss functions reduced the registration error to 1.97 and 4.96. Noticed from Table IV, the single-step registration approach registers the pair of images while reporting higher errors than the methods that use our proposed iterative formulation. Fig. 7 shows how the iterative refinement affects the moving image to be registered with the fixed image. Fig. 7(i)–(k) and (l)–(n) shows the warped image in each iteration by the proposed NBR-Net registration results and corresponding semantic mask. As the

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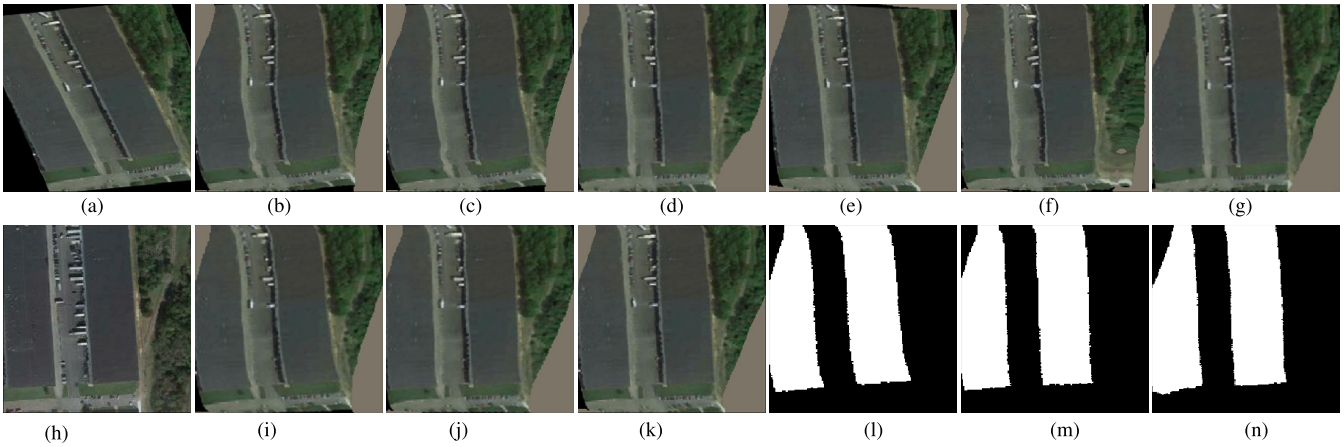


Fig. 7. Visualization of the effect of iterative refinement. (a) Moving image. (h) Fixed image. (b)–(d) Iterative warped image only with the iterative refinement. (e)–(g) Iterative warped image with the iterative refinement and multiple loss functions. (i)–(k) and (l)–(n) Warped images and their labels by our proposed NBR-Net method. (a) Image B. (b) Only Iter1: 5.1. (c) Only Iter2: 3.5. (d) Only Iter3: 2.9. (e) All loss Iter1: 5.1. (f) All loss Iter2: 3.7. (g) All loss Iter3: 2.8. (h) Image A. (i) NBR-Net Iter1:4.6. (j) NBR-Net Iter2:3.1. (k) NBR-Net Iter3:2.6. (l) Label Iter1. (m) Label Iter2. (n) Label Iter3.

number of iterations increases, the registration performance becomes better, which is more clear from the semantic mask.

3) Loss Function: Comparing models 1 and 2 in Table IV, we can find that adding the IOU loss function improves the registration performance by about 0.4, and adding content loss increases by about 0.2. The same conclusion can be drawn by comparing models 6 and 7. Comparing models 4 and 5, we can find that although the performance improvement is not as obvious as the previous groups of comparative experiments, the combined loss function can also improve the registration performance. Fig. 7 (b)–(d) also shows the visible registration results only with the iterative refinement, corresponding to model 4 in Table IV. Fig. 7(e)–(g) shows the registration results with the IOU and content loss functions, corresponding to model 5 in Table IV. Adding IOU loss and the content loss functions reduces the local distortion of the image, as shown in the comparison between Fig. 7(b)–(d) and Fig. 7(e)–(g).

4) Feature Extraction Backbone: First, we conduct experiments to test how different feature extraction backbones affect the proposed NBR-Net. Fig. 8 shows that the NBR-Net using VGG16 has a better performance than NBR-Net using Resnet34, including lower registration error and faster and more stable convergence. Second, we conduct experiments to compare the feature extractor without fixed parameters during training and testing. Experiments show that the fixed feature extractor gets a better registration performance. This is similar to the compared baselines. Third, we compare the registration error using the state-of-the-art Sem GLU with different feature extraction backbones. Experiments show that the registration performance using VGG as the feature extraction is better than that of Resnet. The authors of [6] and [51] also used VGG as a fixed feature extractor. The quantification results are shown in Table V.

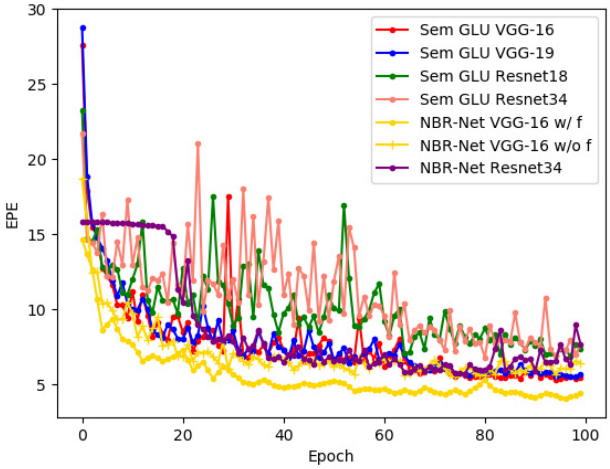


Fig. 8. Convergence curves when training the model using different feature extraction methods.

TABLE V
QUANTITATIVE COMPARISONS WITH DIFFERENT FEATURE EXTRACTION METHODS

	backbone	fixed	EPE
Sem GLU	VGG-16	✓	5.3382
	VGG-19	✓	5.5120
	Resnet18	✓	6.8903
	Resnet34	✓	6.8041
NBR-Net	VGG-16	✓	3.7399
	VGG-16		5.6167
	Resnet34	✓	5.6894

D. UAV Image Registration

In the event of GPS-denied, there is a large translation up to tens of meters, rotation, and scaling displacement for the UAV images. Also, it is inaccessible to retrain or fine-tune models for specific UAV images in emergencies. We believe that the registration model pretrained on multitemporal satellite image dataset will work well.

1) The models extract features with the fixed VGG-16 model and only train the decoder module, alleviating the poor generalization performance caused by inconsistent features.

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2) Bidirectional registration and iterative refinement use stricter constraints to ensure reversibility and consistency of registration at multisteps, reducing performance degradation when dealing with heterogeneous images.
3) Bidirectional registration enhances consistency while maintaining image style, does not involve image style conversion, and avoids content distortion.

Evaluation: We evaluate the image registration performance by the average distance between corresponding keypoints [root mean square error (RMSE)] and the percentage of correct keypoints in n-pixels (PCK-n). We mark 400 and 844 keypoint pairs (building corners, intersections, and landmarks) in the UAV image and the satellite image for evaluation.

The image registration between UAV and satellite images is more challenging and practical. The referenced satellite images and the UAV images in the Village dataset were taken in close time, with obvious texture. Besides comparing with the above-mentioned deep-learning-based methods, we also compare the proposed NBR-Net with some classical image matching methods. However, these traditional methods had a low success rate in the Hunan dataset.

1) SIFT [15]: SIFT is a broadly adopted feature extraction method. The feature is invariant to scale and orientation of images and robust to illumination fluctuations, noise, partial occlusion, and minor viewpoint changes in the images.
2) GMS [25]: The GMS method incorporates the smoothness constraint into a statistic framework for separation and fast feature extraction. GMS is robust to various challenging image changes, involving in viewpoint, scale, and rotation.
3) APAP [56]: This method produces as-projective-as-possible (APAP) image alignment by moving direct linear transformation to fine-tune the projective warp, to accommodate the deviations of the input data from the idealized conditions. This method can accurately align images that differ by more than a pure rotation.
4) Feng [57], [58]: This method first uses SIFT to obtain the global perspective transformation model between images. The registration is then fine-tuned by dividing the image into multiple blocks and improving the relationship between each block using block-weighted projection (BWP) transformation model to map local geometric relations to weighted feature points.
5) Poly [59]: This method presents a novel two-frame motion estimation algorithm, which is an effective optical flow estimation method. The first step is to approximate each neighborhood of both frames by quadratic polynomials, and then to derive the displacement field from the polynomial expansion coefficients.

Table VI reports the numerical UAV image registration performance using several traditional methods. The RMSE error of the unregistered Village UAV image dataset is 7.311, and only less than half of the interest points are within 5 pixels of their corresponding points. SIFT and GMS use a perspective matrix to align the UAV image with the reference image. The SIFT method's registration error is 3.899, but the registration result is greatly affected by parameters. In this article, the flann.knnMatch function is used for matching, the distance ratio of 0.7 and 0.75 has a big gap in matching performance, and the matching errors are 3.899 and 8.916, respectively. Also, only about ten candidate matching pairs can be reserved, which is not enough to support further calculation by grid. The GMS method takes the relative position of candidate matching pairs into consideration, and the matching feature points are concentrated in the center of the image, resulting in large registration errors of other locations. The APAP and Feng methods consider the nonrigid matching and refine the registration results by the grid method. However, they are based on the relatively accurate extraction and matching of SIFT features. Therefore, in the case of less SIFT matching pairs and low matching accuracy, the effect is not even worse than the SIFT method. The registration results of Feng's method are directly used in the BWP-FBM part. APAP and Feng's methods are effective for road registration, but there are many artifacts and errors in other places. Through the communication with the author, we believe that the reasons lie in two aspects: on the one hand, the use scenes are inconsistent, and the UAV image is slightly inclined; on the other hand, the descriptive and robustness of feature extraction methods based on SIFT for satellite and UAV images need to be improved. The Poly method performs well and runs fast, but most image registration fails in multitemporal satellite image dataset and Hunan UAV image dataset, indicating that this method is subject to large image appearance. The similar

TABLE VI

REGISTRATION PERFORMANCE ON VILLAGE IMAGE DATASET USING CLASSICAL METHODS (BOLDED: BEST; UNDERLINED: RUNNER-UP)

	RMSE	PCK-5	Time(s)
Unregistered	7.311	180/400	-
SIFT [15]	3.899	280/400	0.1342
GMS [25]	6.122	257/400	0.5269
APAP [56]	8.161	211/400	2.6301
Feng [57]	7.300	242/400	2.6507
Poly [59]	<u>2.788</u>	<u>330/400</u>	0.0013
NBR-Net	**2.377**	**354/400**	0.3871

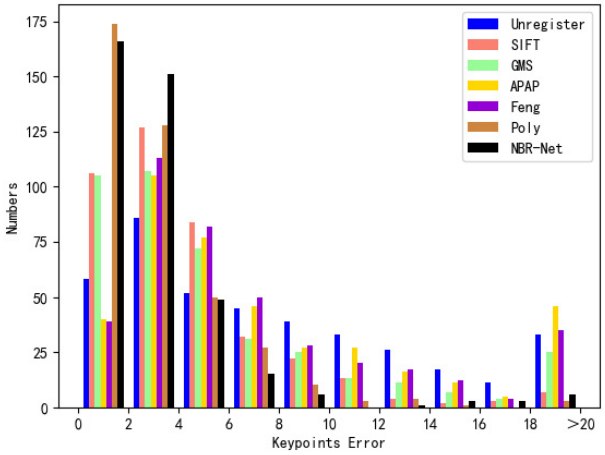


Fig. 9. Statistical registration error of UAV image datasets using traditional classical methods.

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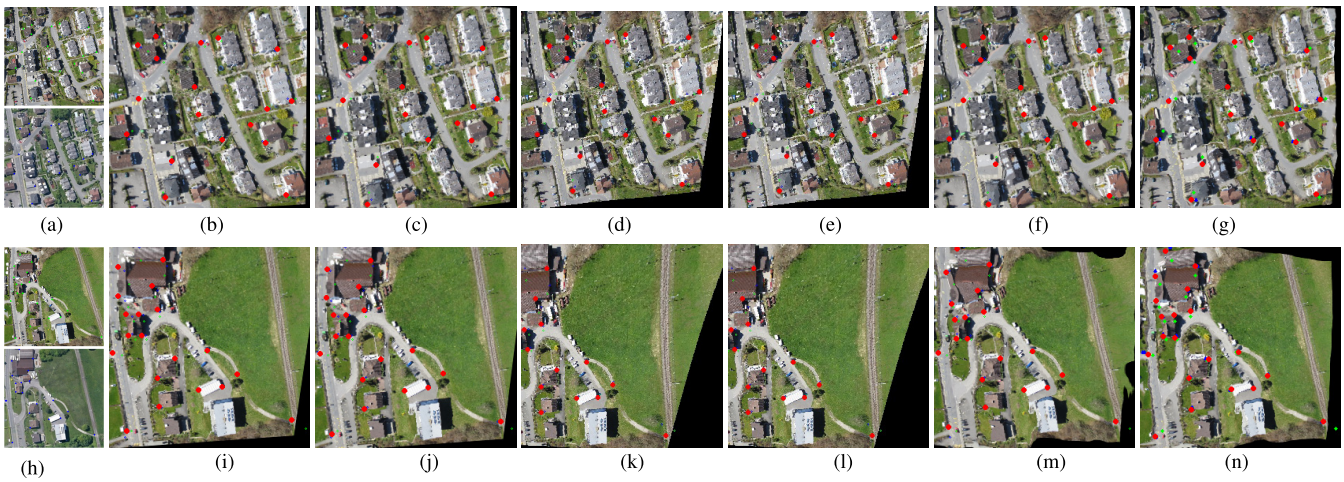


Fig. 10. Visualization of the image registration performance on Village UAV dataset using traditional classical methods. Points in green, blue, and red represent the keypoints in the UAV image, satellite image, and the predicted corresponding points. The first column represents the unregistered image pairs. Top: UAV image. Bottom: satellite image. The registration errors are labeled for quantitative comparison. (a) 6.33. (b) SIFT: 3.77. (c) GMS: 5.81. (d) APAP: 6.97. (e) Feng: 3.52. (f) Poly: 2.22. (g) NBR-Net: 2.22. (h) 7.91. (i) SIFT: 3.78. (j) GMS: 4.40. (k) APAP: 9.65. (l) Feng: 4.53. (m) Poly: 3.53. (n) NBR-Net: 2.97.

TABLE VII
REGISTRATION PERFORMANCE ON UAV IMAGE DATASETS USING DEEP-LEARNING METHODS (BOLDED: BEST; UNDERLINED: RUNNER-UP)

	Village Dataset		Hunan Dataset		Time(s)
	RMSE	PCK-5	RMSE	PCK-10	
Unregistered	7.311	180/400	37.49	2/844	-
CNN-R [26]	5.508	258/400	40.609	189/844	49.6240
ICLK [48]	4.453	259/400	34.337	21/844	1.1506
GLU-Net[6]	4.392	282/400	18.68	412/844	0.0515
GoCor[55]	<u>2.906</u>	<u>336/400</u>	14.82	503/844	0.1363
Sem GLU[6]	3.829	297/400	<u>12.12</u>	<u>618/844</u>	0.1549
NBR-Net	**2.377**	**354/400**	**7.50**	**788/844**	0.3871

pattern can be observed with PCK-5 metric, and the statistical results are shown in Fig. 9. By contrast, our NBR-Net method achieves the best registration result and PCK on all three datasets. The visual registration results of these methods are shown in Fig. 10.

Table VII reports the numerical UAV image registration performance using several deep-learning-based methods and Fig. 11 shows their statistical registration error. The RMSE error of the unregistered Village UAV image dataset is 7.311, and only less than half of the interest points are within five pixels of their corresponding points. Registering by our NBR-Net method, 88.5% of the points of interest are aligned within five pixels of the corresponding points in the satellite image. The original localization of the Hunan UAV image dataset has an uneven offset, and the RMSE registration error is up to 37.5 pixels because of the inaccurate POS information. After registration, there is a significant improvement: 93.4% marked keypoints are aligned within ten pixels of the corresponding points, which is significantly higher than other methods.

Figs. 12 and 13 show the UAV image registration performance using the models pretrained on multitemporal satellite remote sensing image dataset. The appearance between the

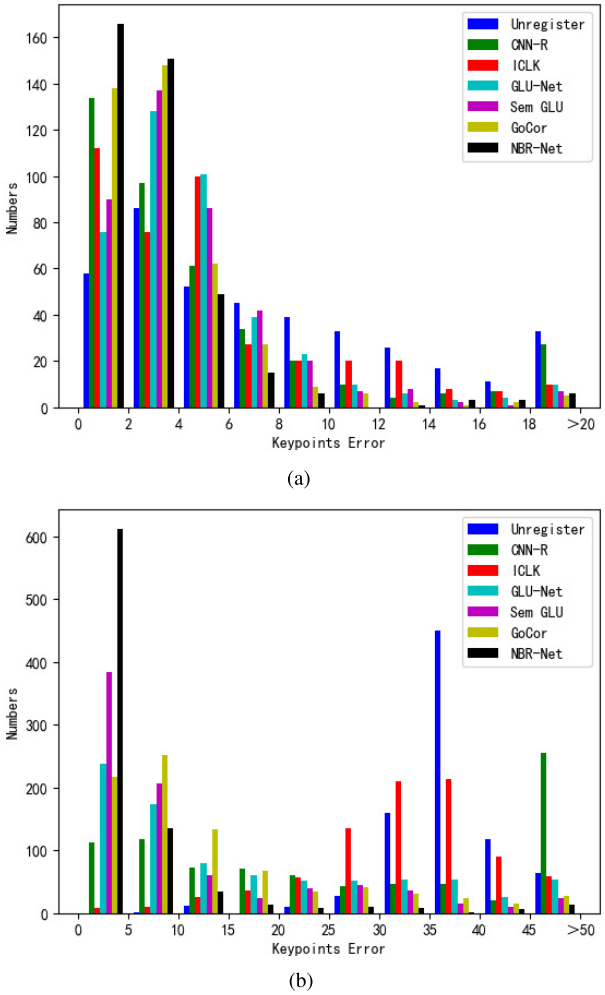


Fig. 11. Statistical registration error of UAV image datasets using deep-learning methods. (a) Village dataset. (b) Hunan dataset.

UAV image and its corresponding satellite image is significantly different and is also different from the multitemporal

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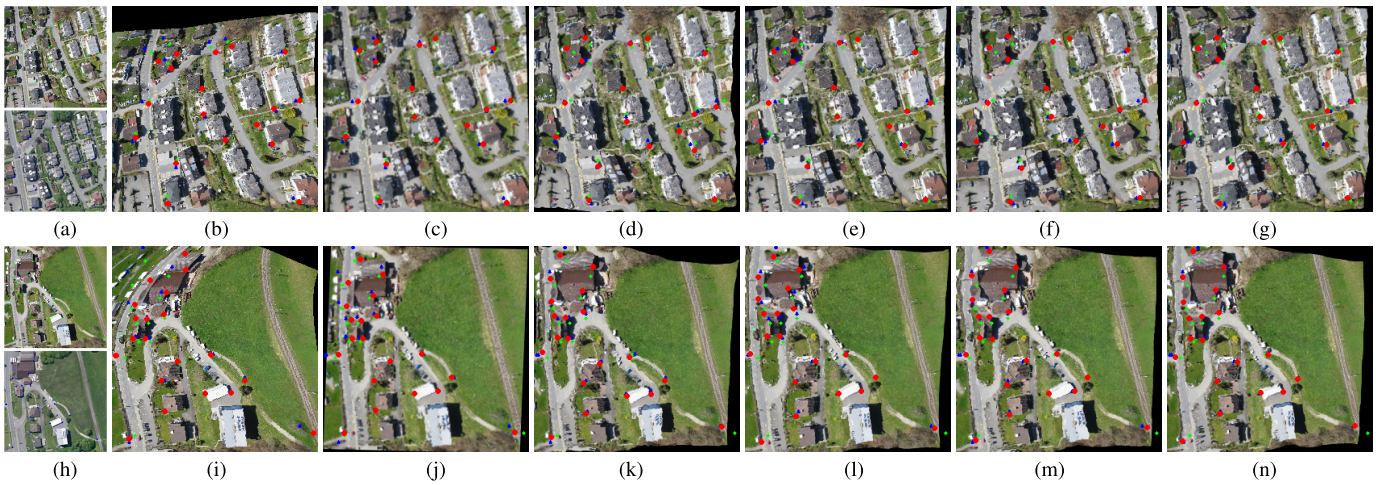


Fig. 12. Visualization of the image registration performance on Village UAV dataset. Points in green, blue, and red represent the keypoints in the UAV image, satellite image, and the predicted corresponding points. The first column represents the unregistered image pairs. Top: UAV image. Bottom: satellite image. The registration errors are labeled for quantitative comparison. (a) 6.33. (b) CNN-R: 8.04. (c) ICLK: 3.76. (d) GLU-Net: 4.11. (e) GoCor: 3.52. (f) Sem GLU: 3.05. (g) NBR-Net: 2.22. (h) 7.91. (i) CNN-R: 7.38. (j) ICLK: 7.54. (k) GLU-Net: 4.53. (l) GoCor: 4.53. (m) Sem GLU: 3.44. (n) NBR-Net: 2.97.

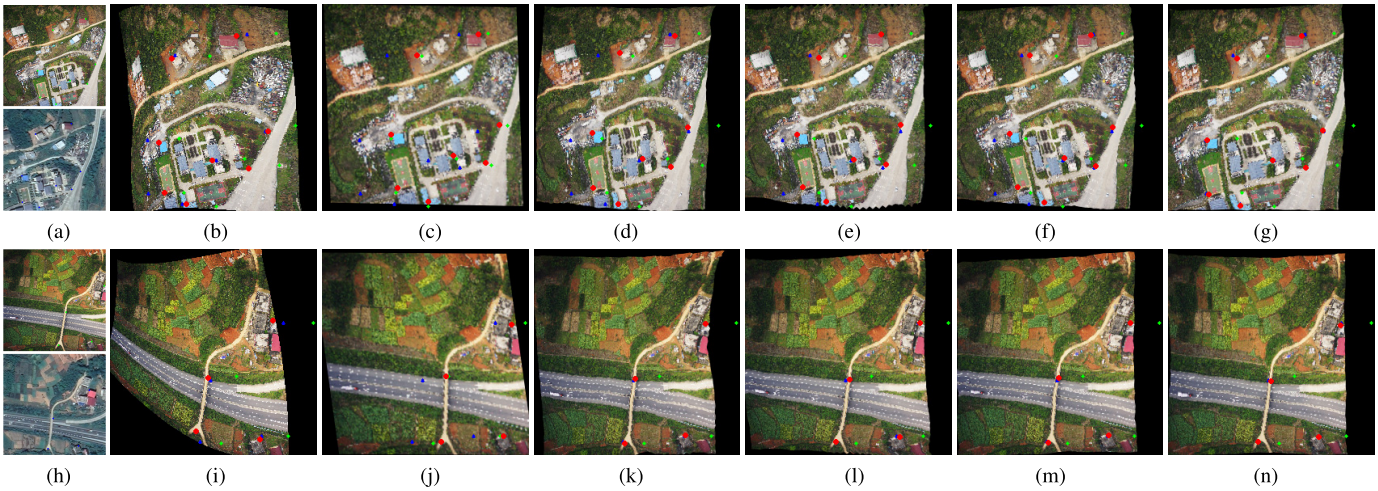


Fig. 13. Visualization of the image registration performance on Hunan UAV dataset. Points in green, blue, and red represent the keypoints in the UAV image, satellite image, and the predicted corresponding points. The first column represents the unregistered image pairs. Up: UAV image. Bottom: satellite image. The registration errors are labeled for quantitative comparison. (a) 39.50. (b) CNN-R: 8.45. (c) ICLK: 35.19. (d) GLU-Net: 14.55. (e) GoCor: 10.65. (f) Sem GLU: 11.69. (g) NBR-Net: 2.68. (h) 32.12. (i) CNN-R: 9.00. (j) ICLK: 30.63. (k) GLU-Net: 2.12. (l) GoCor: 3.06. (m) Sem GLU: 2.06. (n) NBR-Net: 1.93.

satellite image. CNN-R used CNN to generate robust multiscale feature descriptors and matched feature points by EM-based nonrigid transformation. However, the alignment performance is heavily reliant upon the feature matching and even performing worse in Figs. 12(i) and 13(i). Inverse compositional Lucas-Kanade (ICLK) aligned two feature images by an iteratively ICLK layer to optimize the projection matrix. However, the registration performance in the large displacement Hunan image dataset is not satisfactory. Our NBR-Net achieves the best registration performance on these tasks, and the registration accuracy in the center region of the image is higher than that in the edge region. We believe that there are two main reasons for this result: The first is that the center of the image is close to the central viewpoint taken by the UAV, with a smaller offset and stronger consistency. The second is that the texture and details of the middle part are clear, and the relationship between image pixels is relatively accurate. However, many cases far away from the center of the UAV image have no corresponding points in the satellite image, where it is difficult to find the corresponding relationship, resulting in optical flow-field distortion.

Benefiting from the consistency advantage of bidirectional registration and iterative refinement, we can see that the structure of roads and buildings after NBR-Net registration is maintained better, and the registration accuracy is higher than other methods. For example, the structure of the road on the left and the parking lot at the lower-left corner of the first example in Fig. 12 is better maintained. The house in the lower-right corner of the second example in Fig. 12 was initially a regular rectangle but flexibly registered to a less regular one. Moreover, the boundary is maintained well, demonstrating the advantages of nonrigid registration in UAV image registration. Despite the large initial offset of the Hunan dataset, the basketball court at the lower-left corner

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of the first example Fig. 13 is better registered than other methods.

V. Conclusion

This article proposed a nonrigid bidirectional image registration network for remote sensing images and provided a new UAV image registration method in emergencies. For the first time, we propose a bidirectional network by registering Image A to Image B and then back to Image A to strengthen the geometric consistency and reversibility. The intermediate registration results are continuously refined through iterative refinement, improving the registration effect of images with large displacement. We strengthen the loss constraint of bidirectional tasks, multistep iterations, and multiscale layers to maintain image content between remote sensing images with significant appearances and large displacements. Experiments on a multitemporal remote sensing image dataset and two GPS-denied UAV image datasets with significant appearance differences verify the advantages of our method.

In the future, we will focus on the transfer learning of registration with a small number of UAV images and bidirectional registration in a weakly supervised manner.

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Yingxiao Xu received the B.S. degree in information and communication engineering from the National University of Defense Technology, Changsha, China, in 2019, where she is pursuing the Ph.D. degree with the School of Electronic Science.

Her research interests include machine learning, remote sensing image processing, and photogrammetry.



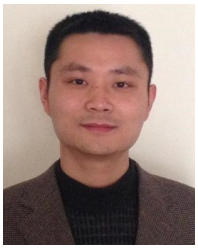
Jun Li received the Ph.D. degree in information and communication engineering from the National University of Defense Technology, Changsha, China, in 2000.

He is a Professor with the National University of Defense Technology. His research interests include management and analysis of big data and spatial information system.



Chun Du received the Ph.D. degree in electronic science and technology from the National University of Defense Technology, Changsha, China, in 2014.

He is an Associate Professor with the National University of Defense Technology. His research interests include machine learning, machine vision, and remote sensing.



Hao Chen received the Ph.D. degree in information and communication engineering from the National University of Defense Technology, Changsha, China, in 2010.

He is a Professor with the National University of Defense Technology. His research interests include data mining, machine learning, and evolutionary computation.

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